Cells and Gauge

File: gauge.kicad_sch

Sleep Current and 3V3

File: power.kicad_sch

MCU and NTCs

File: mcu.kicad_sch

USB Port 1

IRQD IRQ1
EN_5V EN_5V1
ADCIN1 P1_ADCIN1
ADCIN2 P1_ADCIN2
LDO_3V3D P1_LDO_3V3

File: usb_port.kicad_sch

USB Port 2

IRQD IRQ2
EN_5V EN_5V2
ADCIN1 P2_ADCIN1
ADCIN2 P2_ADCIN2
LDO_3V3D

File: usb_port.kicad_sch

P1_LDO_3V3
R1 0R
P1_ADCIN1
R2 0R
P1_ADCIN2
GND

P2_ADCIN1
R3 0R
P2_ADCIN2
R4 0R
GND

TPS25751 ADDRESS / DEAD-BATTERY STRAPS (SafeMode)
Port 1: R1 ADCIN1 = LDO_3V3, R2 ADCIN2 = GND  -> I2C 0x20
Port 2: R3 ADCIN1 = GND,     R4 ADCIN2 = GND  -> I2C 0x21
Gauge BQ40Z50-R2 = 0x0B on the same SDA/SCL bus (STM32WB I2C1, 100 kHz)

MONOLITH DEV BOARD - SHEET INDEX
2  Cells and Gauge        4S holders / pack input, BQ40Z50-R2, protection FETs, SMBus
3  Sleep Current and 3V3  R_MEAS + 2x INA190 to SMA, TPS70950 amp supply, TPS62177 3V3
4  MCU and NTCs           STM32WB1MMC, debug headers, board NTCs TH1-TH4, LED
5  USB Port 1 / 6  USB Port 2   (same sheet file)  TPS25751D + BQ25756 + TPS62933 per port
Rails (power symbols): PACK_P, 3V3, GND, CELL_N (gauge VSS only)
Spec: docs/devboard/circuit-description.md   Changes/open items: DevBoard/README.md

REVISION LOG
A0  2026-10-11  First draft from circuit description draft 5.  Author: Claude Code.  Reviewer: none yet.
A1  2026-10-11  Layout cleanup: blocks, power symbols, notes, title blocks.  Author: Claude Code.  Reviewer: none yet.
A2  2026-10-11  Readability pass: no overlapping text, sheet-pin names inside boxes.  Author: Claude Code.  Reviewer: none yet.
A3  2026-10-11  Wiring pass: wires instead of labels inside each block.  Author: Claude Code.  Reviewer: none yet.
A4  2026-10-12  Polymer bulk caps, PWR_FLAGs at rail sources, design-review fixes
                (docs/devboard/schematic-review-2026-10-12.md).  Author: Claude Code.  Reviewer: 3-agent review.
A5  2026-10-13  Review 2 decisions: CHG_INH fails safe (R417/R517 47k pull-up to 3V3, MCU drives low to charge),
                MCU IWDG in hardware mode; BAT fed before F101 (TI fig. 9-12); C421/C521 100 uF; LRB520S, BSS138,
                C109 4.7n, PPHV/VBUS caps, GND/probe test points, JLC numbers on all R/C (review 2).  Author: Claude Code.

Not reviewed, laid out, built or tested
Spec + open items: DevBoard/README.md
Sheet index, PD address straps
**Monolith 100 W battery bank — dev board**

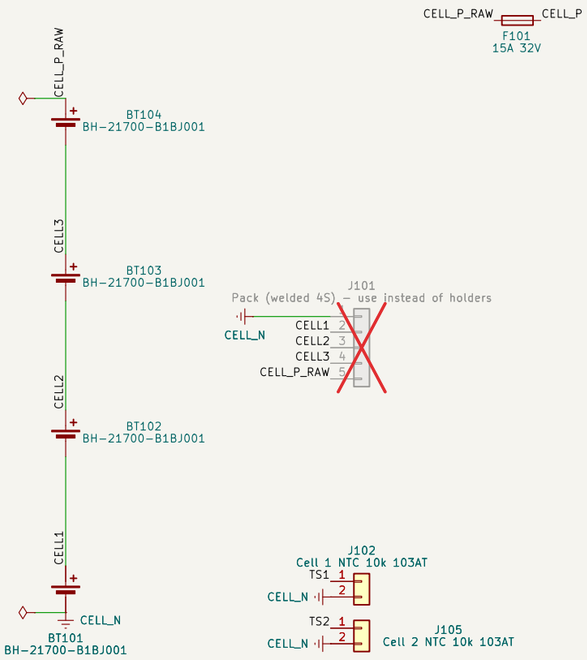
Sheet: /
File: MonolithDev.kicad_sch

***Title: Monolith dev board — top level***

| Size: A3 | Date: 2026-10-11 | **Rev: A (draft)** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 1/6 |

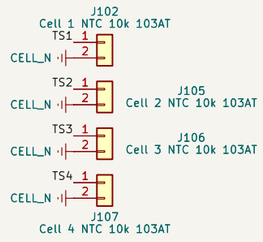
## CELL STACK (4S)

*Holders BT101–BT104 OR welded pack on J101 — never both.*
*J101 is DNP. Check each cell at TP102–TP105 and the stack at TP108 before waking the gauge (Q101/Q102 stay off until FET_EN is set, so nothing loads the cells until then).*

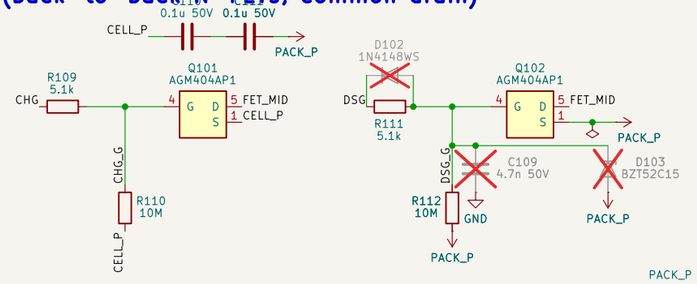




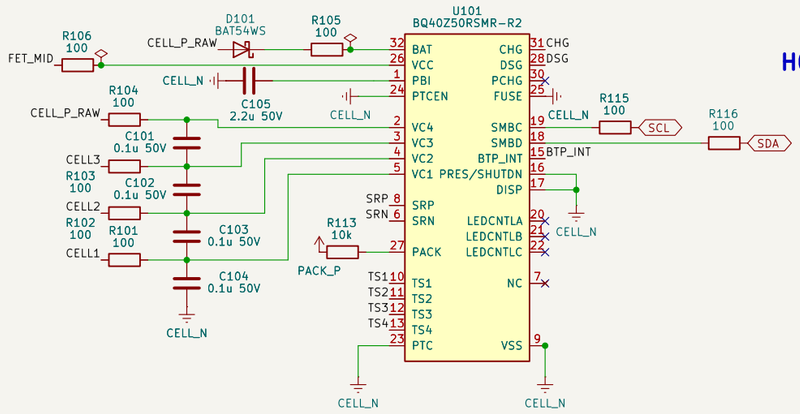




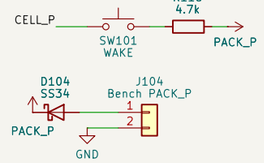

## PROTECTION FETs (back-to-back N-FETs, common drain)



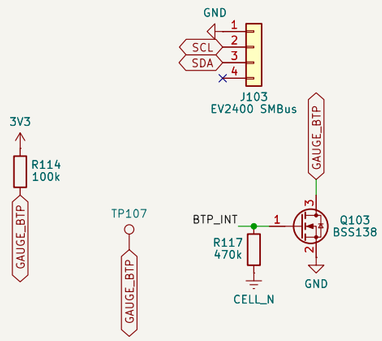

## GAS GAUGE BQ40Z50-R2 (TI fig. 9-1)



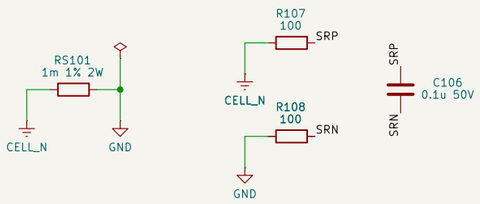

## WAKE + BENCH SUPPLY




## HOST SMBus (gauge 0x0B) / EV2400




## LOW-SIDE CURRENT SENSE




## TEST POINTS / POWER FLAGS

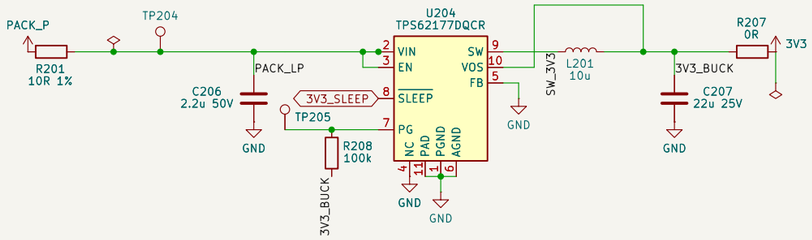



NOTES
1. CELL_N = gauge VSS (cell 1N tab). GND = system ground, on the far side of RS101.
2. Kelvin-route SRP/SRN from the RS101 pads; R107/R108/C106 at the IC.
3. VC taps are separate thin traces from the cell end of each holder contact;
   VC4 from CELL_P_RAW (before F101) so fuse drop is not read as cell voltage.
4. TS1–TS4: one wired cell NTC each (J102, J105–J107, one per cell); insulated sensors only (cans 0–12.6 V above VSS, TS 2.1 V max). PTC, PTCEN, PRE
5. VCC fed from FET common drain via R106 (TI R13). PACK sense via R113 10k (TI R15).
6. BAT fed from CELL_P (after fuse, before FETs) via D101 + R105.
7. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
8. Q103 buffers BTP_INT (its high level has no maximum; PA6 takes 4 V unpowered): GAUGE_BTP is active low.
9. C109 + D102 (DNP): fit if the DSG turn-on inrush into PACK_P trips SCD. D103 (DNP) clamps Q102 VGS in a hard short.
10. Tape the J102 NTC to BT101: a 21700 can is its cell's negative terminal and TS1 is rated 2.1 V.
11. SW101 wakes the gauge from SHUTDOWN without USB; J104 feeds PACK_P from a current-limited bench supply.
12. J103 matches the EV2400 SMBus port: 1 VSS, 2 SCL, 3 SDA, 4 VOUT (not used). Fit cells before plugging in USB.
13. ESD caps (TI): C110+C111 in series across the FET pair (fig. 9-2 C1/C2). C107+C108 across PACK+/PACK− (sec. 9.2.2.1.5).

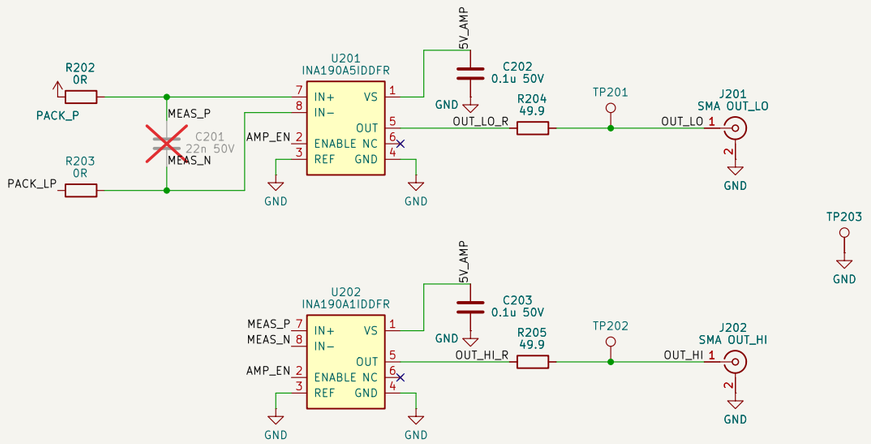
Not reviewed, laid out, built or tested
Spec + open items: DevBoard/README.md
4S input, BQ40Z50-R2 gauge/protector, protection FETs
**Monolith 100 W battery bank — dev board**

| | | |
|---|---|---|
| Sheet: /Cells and Gauge/ | | |
| File: gauge.kicad_sch | | |
| ***Title: Cells and gauge*** | | |
| Size: A3 | Date: 2026-10-11 | **Rev: A (draft)** |
| KiCad E.D.A. 10.0.7 | | Id: 2/6 |

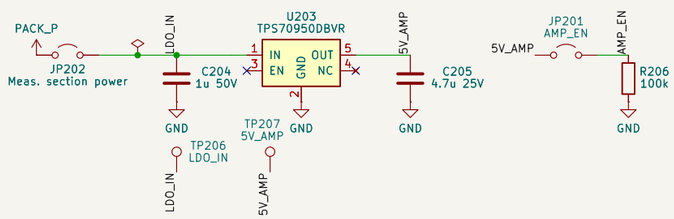
## 3.3 V ALWAYS-ON RAIL  (TPS62177, 1 MHz, sleep mode under MCU control)




## SLEEP-CURRENT MEASUREMENT  (both amps across R201 = R_MEAS)




## AMPLIFIER SUPPLY  (5 V LDO, fed from PACK_P so amp current bypasses R_MEAS)




NOTES
1. Low-current branch: PACK_P -> R201 10R -> PACK_LP -> TPS62177 -> 3V3 (MCU + both TPS25751).
   Gauge BAT is NOT on this branch (TI: gauge must stay powered with FETs off).
2. Kelvin-connect R202/R203 to the R201 pads.
3. U201 gain 500: 5 mV per uA, -0.99 mA full scale.  U202 gain 25: 0.25 V per mA, -20 mA full scale.
4. R202/R203/C201 fitted 0R/0R/DNP to start; 100R/100R/22nF gives a 36 kHz input filter.
5. R204/R205 49.9R isolate the INA190 outputs from cable capacitance (1 nF max); use a 1 MOhm scope input.
6. SLEEP low = sleep mode (internal pull-down); MCU PB4 drives it high for full load.
7. R207 breaks the 3V3 rail for debugging.
8. U201/U202 are hand-soldered (DigiKey/Mouser); JLC has no stock. Not in the JLC BOM/CPL.
9. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
10. U203 EN is left open (internal 300 nA pull-up enables it). EN must stay below 6.5 V: never tie it to LDO_IN.
11. U204 pin 4 (NC) goes to GND, as TI recommends.

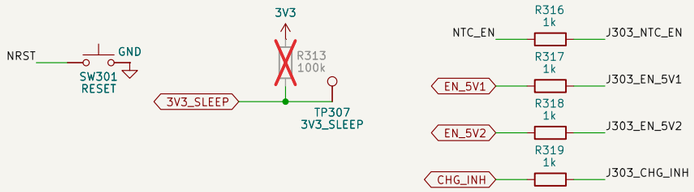
Not reviewed, laid out, built or tested
Spec + open items: DevBoard/README.md
R_MEAS + INA190 x2, TPS70950, TPS62177
**Monolith 100 W battery bank — dev board**

Sheet: /Sleep Current and 3V3/
File: power.kicad_sch

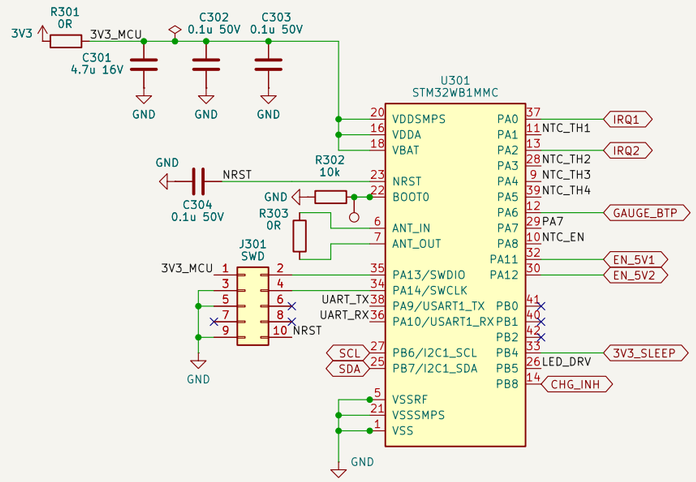
***Title: Sleep-current measurement and 3.3 V rail***

| Size: A3 | Date: 2026-10-11 | **Rev: A (draft)** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 3/6 |

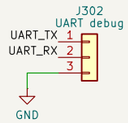
## BRING-UP AIDS




## BLUETOOTH MCU  STM32WB1MMC  (per ST DS14096 fig. 5)



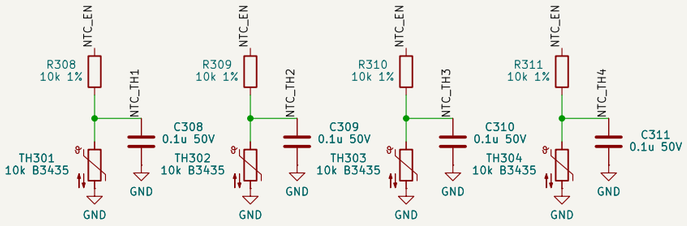

## DEBUG HEADERS



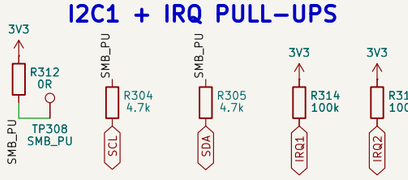




## BOARD NTCs  (TH1–TH4, divider powered only while sampling)

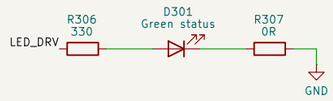



## I2C1 + IRQ PULL-UPS




NOTES
1. U301 at a board edge, as on ST's reference PCB (DS14096 fig. 6): copper-free notch past the antenna end.
2. C301–C303 at the VDD pads; C304 at NRST.
3. PB0/PB1/PB2 left unconnected (ST: RF-sensitive pins). NTC_EN = PA8, EN_5V1 = PA11, EN_5V2 = PA12.
4. R303 0R joins ANT_IN/ANT_OUT for the internal antenna (ST fig. 2).
5. Pin map checked against DS13258 table 15: NTC ADC1_IN6/8/9/10, IRQs on WKUP1/WKUP4, I2C1, USART1.
   PB4 = 3V3_SLEEP: PB4 resets as NJTRST with pull-up, so the 3V3 rail boots in normal mode;
   firmware makes PB4 a GPIO output before using sleep mode. PB5 = LED, PA6 = GAUGE_BTP.
6. R307 removes the LED for sleep-current measurements.
7. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
8. SW301 resets the MCU. TP306 is BOOT0: hold it at 3V3 while releasing reset for the UART bootloader.
9. R313 (DNP) holds 3V3_SLEEP high (normal mode) while the MCU is off. TP307 probes it.
10. R312 feeds the SMBus pull-ups: remove it and supply TP308 for EV2400 work while 3V3 is down.
11. R316–R319: J303 can override NTC_EN/EN_5V1/EN_5V2/CHG_INH without shorting the MCU outputs.
12. PB8 = CHG_INH (rev A5): drive LOW to allow charging. High or hi-Z stops both chargers (R417/R517 pull CE up to 3V3), so reset/hang/unprogrammed stops charging. Low only while charging (69 uA per port), high in idle. No Standby.
13. Watchdog: IWDG in hardware mode (option byte IWDG_SW = 0) restarts a hung MCU; it runs from LSI, also in Stop/Standby.

## STATUS LED







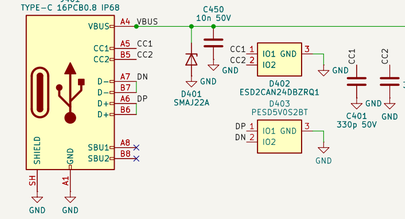
Not reviewed, laid out, built or tested
Spec + open items: DevBoard/README.md
STM32WB1MMC, debug headers, NTC TH1–TH4
**Monolith 100 W battery bank — dev board**

Sheet: /MCU and NTCs/
File: mcu.kicad_sch

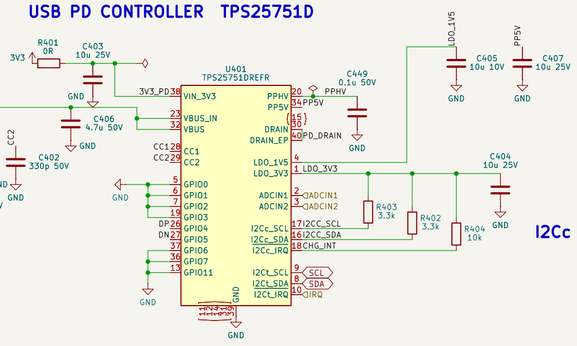
***Title: MCU and board NTCs***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-11 | **Rev: A (draft)** |
| KiCad E.D.A. 10.0.7 | | Id: 4/6 |

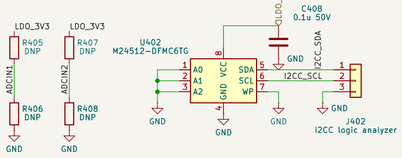
## USB-C CONNECTOR + PROTECTION



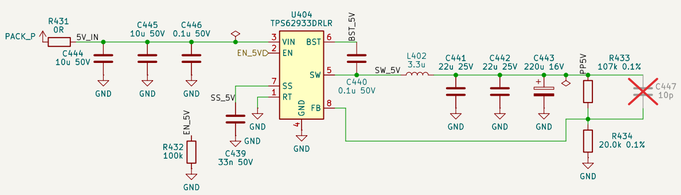

## USB PD CONTROLLER TPS25751D



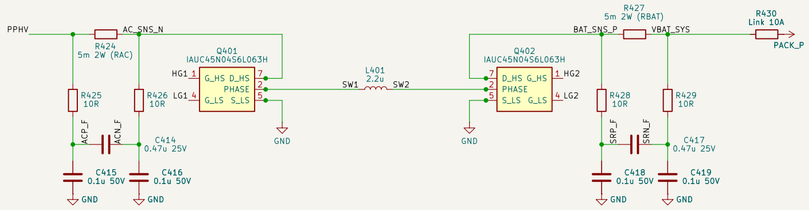

## I2Cc BUS: EEPROM + PULL-UPS




## PP5V SUPPLY TPS62933 (1.2 MHz, 5.08 V) — 5 V source + VCONN

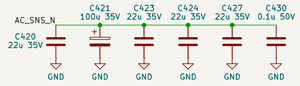



## BQ25756 4-SWITCH BUCK-BOOST POWER STAGE (copy production layout exactly)




USB-SIDE BULK: 4 x 22u MLCC + 68u polymer (>= 80 uF effective)

BATTERY-SIDE BULK: 3 x 47u MLCC + 100u polymer (>= 80 uF effective)

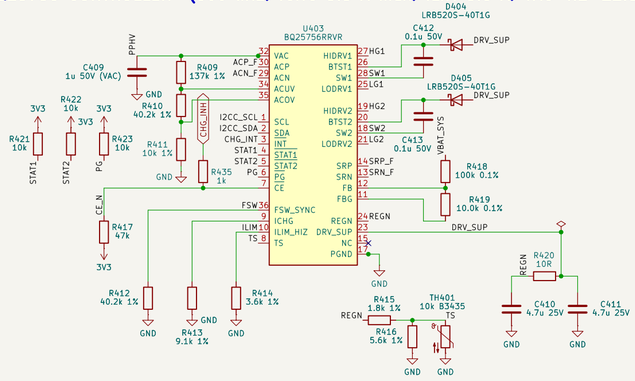






NOTES (this sheet is used twice: USB Port 1 = 4xx, USB Port 2 = 5xx)
1. Address straps are on the root sheet: port 1 ADCIN1 = LDO_3V3, ADCIN2 = GND (0x20); port 2 ADCIN1 = ADCIN2 = GND (0x21). R405–R408 are DNP divider footprints for other options.
2. Unused TPS25751 GPIOs tied to GND per datasheet (KiCad ERC flags this; expected).
3. At the connector pins: D402 CC ESD (24 V). D403 D+/D− ESD (5 V; GPIO4/5 are rated 6 V). C401/C402.
4. Kelvin-route ACP/ACN from R424 and SRP/SRN from R427; filter parts at the IC.
5. Two MLCCs of each bulk bank plus C430/C438 at the FETs; the polymer (C421/C432) and remaining MLCCs one row back. Polymer carries the bulk (no DC-bias loss). MLCCs carry the switching-frequency ripple.
6. I2Cc pull-ups go to this TPS25751's own LDO_3V3 (TI guidance).
7. R430 = 10 A copper link to PACK_P (net-tie footprint, not a part): widen with zones.
8. VAC is taken before R424 per TI typical application; DRV_SUP fed from REGN via R420.
9. PPHV bulk (BQ25756 input bank) exceeds the 100 uF USB PD c5xxBulkPd: the TPS25751D sink switch soft-start limits inrush (C x SS: about 125 uF effective x 0.47 V/ms max = 0.06 A at SS setting 0). Set PPHV soft-start to setting 0 or 1 in the TPS25751 configuration.
10. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
11. CHG_INH (rev A5): R417 pulls CE up to 3V3, so charging needs the MCU to drive CHG_INH low through R435. Reset, IWDG reset, a hang or an unprogrammed MCU stop charging; a dead pack (3V3 down) still charges. Keep DIS_CE_PIN = 0.

## BQ25756 CONTROLLER (600 kHz, ICHG 5.5 A max, ILIM 5.6 A, VAC 4.1–22.5 V window)




## TEST POINTS / POWER FLAGS



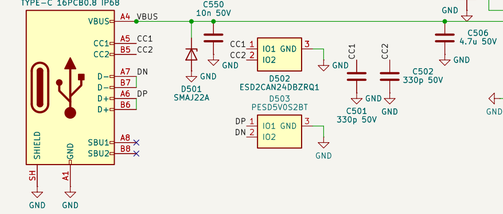

Not reviewed, laid out, built or tested
Spec + open items: DevBoard/README.md
TPS25751D + BQ25756 + TPS62933; port 1 = 4xx, port 2 = 5xx
**Monolith 100 W battery bank — dev board**

Sheet: /USB Port 1/
File: usb_port.kicad_sch

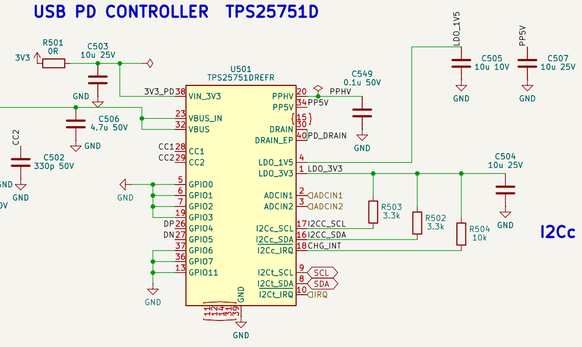
**Title: USB-C PD port (used twice)**

| Size: A2 | Date: 2026-10-11 | **Rev: A (draft)** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 5/6 |

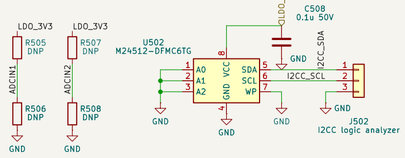
## USB-C CONNECTOR + PROTECTION



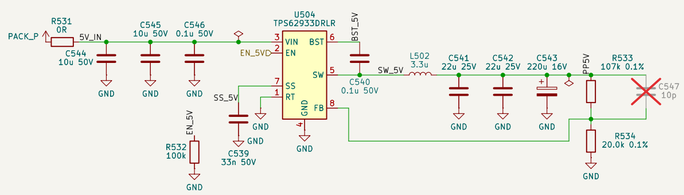

## USB PD CONTROLLER TPS25751D



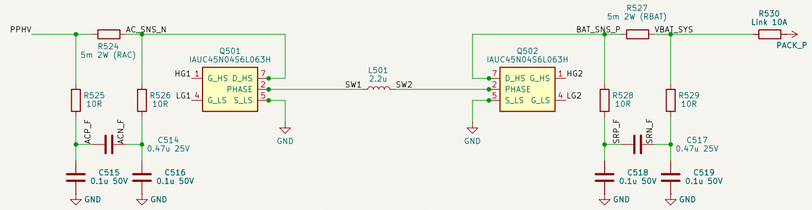

## I2Cc BUS: EEPROM + PULL-UPS




## PP5V SUPPLY TPS62933 (1.2 MHz, 5.08 V) — 5 V source + VCONN

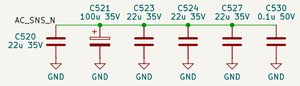



## BQ25756 4-SWITCH BUCK-BOOST POWER STAGE (copy production layout exactly)




USB-SIDE BULK: 4 x 22u MLCC + 68u polymer (>= 80 uF effective)

BATTERY-SIDE BULK: 3 x 47u MLCC + 100u polymer (>= 80 uF effective)

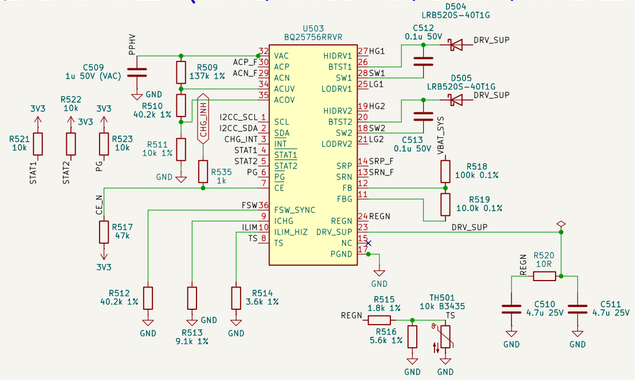






NOTES (this sheet is used twice: USB Port 1 = 4xx, USB Port 2 = 5xx)
1. Address straps are on the root sheet: port 1 ADCIN1 = LDO_3V3, ADCIN2 = GND (0x20); port 2 ADCIN1 = ADCIN2 = GND (0x21). R405–R408 are DNP divider footprints for other options.
2. Unused TPS25751 GPIOs tied to GND per datasheet (KiCad ERC flags this; expected).
3. At the connector pins: D402 CC ESD (24 V). D403 D+/D− ESD (5 V; GPIO4/5 are rated 6 V). C401/C402.
4. Kelvin-route ACP/ACN from R424 and SRP/SRN from R427: filter parts at the IC.
5. Two MLCCs of each bulk bank plus C430/C438 at the FETs; the polymer (C421/C432) and remaining MLCCs one row back. Polymer carries the bulk (no DC-bias loss). MLCCs carry the switching-frequency ripple.
6. I2Cc pull-ups go to this TPS25751's own LDO_3V3 (TI guidance).
7. R430 = 10 A copper link to PACK_P (net-tie footprint, not a part): widen with zones.
8. VAC is taken before R424 per TI typical application; DRV_SUP fed from REGN via R420.
9. PPHV bulk (BQ25756 input bank) exceeds the 100 uF USB PD c5xxBulkPd: the TPS25751D sink switch soft-start limits inrush (C x SS: about 125 uF effective x 0.47 V/ms max = 0.06 A at SS setting 0). Set PPHV soft-start to setting 0 or 1 in the TPS25751 configuration.
10. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
11. CHG_INH (rev A5): R417 pulls CE up to 3V3, so charging needs the MCU to drive CHG_INH low through R435. Reset, IWDG reset, a hang or an unprogrammed MCU stop charging; a dead pack (3V3 down) still charges. Keep DIS_CE_PIN = 0.

## BQ25756 CONTROLLER (600 kHz, ICHG 5.5 A max, ILIM 5.6 A, VAC 4.1–22.5 V window)




## TEST POINTS / POWER FLAGS




Not reviewed, laid out, built or tested
Spec + open items: DevBoard/README.md
TPS25751D + BQ25756 + TPS62933; port 1 = 4xx, port 2 = 5xx
**Monolith 100 W battery bank — dev board**

Sheet: /USB Port 2/
File: usb_port.kicad_sch

***Title: USB-C PD port (used twice)***

| Size: A2 | Date: 2026-10-11 | **Rev: A (draft)** |
|---|---|---|
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